

The Rosen Fibration

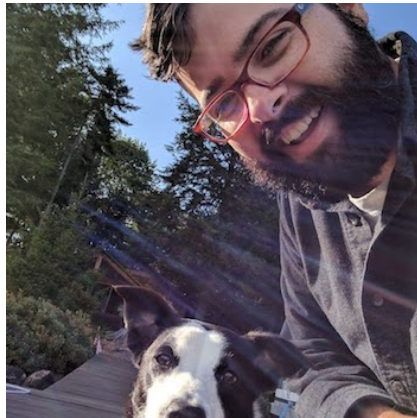
ACT26 — Tallinn

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Tallinn University of Technology / SUNY Binghamton

How it started

At the end of October 2025, Amahury (L) and Sean (R, with doggo) contacted me.



- Applying category theory to biology: 'vindicating' Robert Rosen, who believed category theory was a foundation for life sciences.

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- This is a talk about mathematics.

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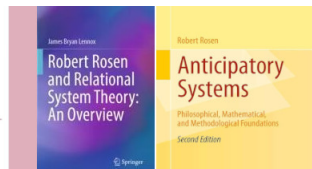


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- Wrote extensively on category theory in biology
- not always correct...

Wrote **before** Yoneda lemma, adjunctions, monads ... and stayed outside the category theory community.

ESSAYS
ON
LIFE
ITSELF

ROBERT ROSEN



MR (or '(M,R)') systems

A 'living system' is capable of
M) etabolism

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A **candle** has f but not Φ ; **Ni-Ti alloy** has Φ (seals microcracks) but no metabolism.

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core idea: there is an intrinsic self-referentiality property for alive systems (“closure under efficient causation”: $\exists \bar{a}. \lambda x. \Phi(f\bar{a})x = \lambda x. fx$) but lacks monoidal closed categories (1958).

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Our goal: study the diagram $A \rightarrow B \rightarrow [A, B]$ in a suitable sense. Can we build a category

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Yes — **MRS** is the **Rosen fibration** of the title.

Incoherent systems

Working Assumption

$\mathcal{C}$ is symmetric monoidal closed, with all needed (co)limits.

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Definition

An **incoherent MR system** is $A \xrightarrow{f} B \xrightarrow{\Phi} [A, B]$ in $\mathcal{C}$. A morphism $(\ell, r) : (f, \Phi) \rightarrow (g, \Psi)$ makes

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\ \downarrow l & & \downarrow r & & \searrow [A, r] \\ A' & \xrightarrow{g} & B' & \xrightarrow{\Psi} & [A', B'] \\ & & & & \nearrow [\ell, B'] \\ & & & & [A, B'] \end{array}$$

commute. The total category **iMRS** has triples $(A, B; f, \Phi)$ as objects and morphisms as above.

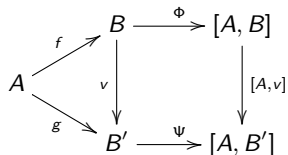
I use agda, btw

```
1 record iMR2 (A B : Obj) : Set (o ∪ ℓ) where
2   constructor ⟨_,_⟩
3   field
4     f : A ⇒ B
5     φ : B ⇒ [ A , B ]o
6
7 record iMR2o : Set (o ∪ ℓ) where
8   field
9     A B : Obj
10    ξ : iMR2 A B
11
12 record iMR2⇒ (X Y : iMR2o) : Set (o ∪ ℓ ∪ e) where
13   module X = iMR2o X
14   module Y = iMR2o Y
15   module ξX = iMR2 X.ξ
16   module ξY = iMR2 Y.ξ
17   field
18     l : X.A ⇒ Y.A
19     r : X.B ⇒ Y.B
20     eqf : r ∘ ξX.f ≈ ξY.f ∘ l
21     eqφ : [ l , id ]1 ∘ ξY.φ ∘ r ≈ [ id , r ]1 ∘ ξX.φ
```


Reasoning fiberwise

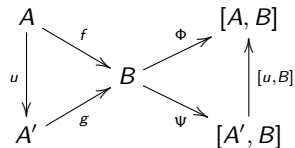
($\hookrightarrow$) Fix A : category $\mathbf{iMRS}^A$ ($\ell = \text{id}$)

- same A , varying B ,
- morphisms $v : B \rightarrow B'$ with



($\hookrightarrow$) Fix B : category $\mathbf{iMRS}_B$ ($r = \text{id}$)

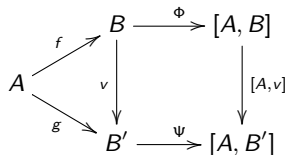
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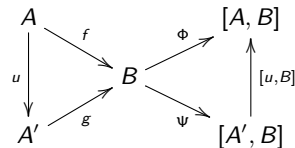
```

1 record iMR2A (A : Obj) : Set (o u ℓ u e) where
2   field
3     B : Obj
4     ξ : iMR2 A B
5
6 record iMR2A (A : Obj) (X Y : iMR2A A) : Set (o u ℓ u e) where
7   module X = iMR2A X
8   module Y = iMR2A Y
9   module EX = iMR2 X.ξ
10  module EY = iMR2 Y.ξ
11  field
12    v : X.B → Y.B
13    eqf : v ∘ EX.f = EY.f ∘ u
14    eqφ : [ id , v ] ∘ EX.φ = EY.φ ∘ v

```

(☞) Fix B : category $\mathbf{iMRS}_B$ ($r = \text{id}$)

- same B , varying A ,
- morphisms $u : A \rightarrow A'$ with



```

1 record iMR2B (B : Obj) : Set (o u ℓ u e) where
2   field
3     A : Obj
4     ξ : iMR2 A B
5
6 record iMR2B (B : Obj) (X Y : iMR2B B) : Set (o u ℓ u e) where
7   module X = iMR2B X
8   module Y = iMR2B Y
9   module EX = iMR2 X.ξ
10  module EY = iMR2 Y.ξ
11  field
12    u : X.A → Y.A
13    eqf : EX.f ∘ u = EY.f ∘ u
14    eqφ : EX.φ ∘ [ u , id ] = EY.φ

```

Incoherent systems

Equivalent characterization of an incoherent MR system:

- Fix A ; one can think of the category of iMRS 'based at A ' as the category of algebras for the endofunctor of $\mathcal{C}$

$$X \mapsto A \otimes (I + X)$$

- Fix B ; the category of iMRS 'based at B ' is the slice $\mathcal{C}/[I + B, B]$.

A slightly more refined characterization, that works only in a distributive Cartesian base, is the following:

- An iMR based at A is a parametric, pointed dynamical system of type $\xi \approx (1 \xrightarrow{f} B; B \xrightarrow{\phi} B)$ with certain coKleisli maps;
- More simply, an iMR seems to be a **lens** $(\frac{A}{B}) \rightarrow (\frac{B}{B})$. (There is more to this story I haven't looked into...)

There is a problem!

() $A \mapsto \mathbf{iMRS}^A$ is functorial (reindexing $v^* : \mathbf{iMRS}^{A'} \rightarrow \mathbf{iMRS}^A$).

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$(\text{reindex}) A \mapsto \mathbf{iMRS}^A$ is functorial (reindexing $v^* : \mathbf{iMRS}^{A'} \rightarrow \mathbf{iMRS}^A$).

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```
1 left : {u : A → A'} → Functor (IMR2A A') (IMR2A A)
2 left {A} {A'} u = record
3   { Fv = λ { x →
4     let module x = IMR2A x
5     in record
6       { B = x.B
7       ; ξ = λ IMR2.f x.ξ → u , [ u , id ]₁ • IMR2.φ x.ξ }
8     }
9   ; Fv = λ { {x} {y} f →
10     let module x = IMR2A x
11     module ξx = IMR2 x.ξ
12     module y = IMR2A y
13     module ξy = IMR2 y.ξ
14     module f = IMR2A f
15     in record
16       { v = f.v
17       ; eqf = begin f.v • f.ξx.f • u = ( sym-assoc )
18         ( f.v • f.ξx.f ) • u = ( f.eqf ) • (refl )
19         f.ξy.f • u
20       ; eqφ = begin [ id , f.v ]₁ • [ u , id ]₁ • f.ξx.φ = ( sym-assoc • ([ [-,-] ]-commute ) • (refl) • assoc )
21         [ u , id ]₁ • [ id , f.v ]₁ • f.ξx.φ = ( (refl) • ( f.eqφ ) • sym-assoc )
22         ([ u , id ]₁ • f.ξy.φ ) • f.v
23       }
24   ; identity = λ {A} → refl
25   ; homomorphism = λ {X} {Y} {Z} {f} {g} → refl
26   ; Fresp = λ x → x
27 }
28
```

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any mapping $t : X \rightarrow Y$ into the same mapping t of Set. The replication maps Φ_f are now defined by the *natural transformations* $\Phi : I \rightarrow h^X$, with X varying in $\mathbf{M}$. Let us denote by (h^X, I) -the sets of all natural transformations from I to h^X .

Coherent MR systems

Φ is pointed by f ; naturality means

commutes whenever $\begin{array}{ccc} X & \xrightarrow{h} & Y \\ \ell \downarrow & & \downarrow r \\ X' & \xrightarrow{k} & Y' \end{array}$ is an arrow $h \rightarrow k$ in the category $\mathcal{C}^{[0 \rightarrow 1]}$.

$$\begin{array}{ccc} Y & \xrightarrow{\Phi_h} & [A, Y] \\ r \downarrow & & \downarrow [A, r] \\ Y' & \xrightarrow{\Phi_k} & [A, Y'] \end{array}$$

Towards the Rosen fibration

The profunctor of MR systems | 

MRS(A, B) is a profunctor $\mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathbf{Set}$; its action on morphisms $u : A' \rightarrow A$, $v : B \rightarrow B'$ is

$$\begin{aligned} \mathbf{MRS}(A, B) &\longrightarrow \mathbf{MRS}(A', B') \\ (A \rightarrow B, \Phi) &\longmapsto (A' \rightarrow A \rightarrow B \rightarrow B', [u, \text{cod}] \circ \Phi) \end{aligned}$$

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```
1 MRS-Profunctor : Bifunctor (Category.op C) C (Setoids (o u l u e) (o u l u e))
2 MRS-Profunctor = record
3   { F0 = λ { (A , B) → MR2-Setoid A B }
4   ; F1 = λ { {(A , B)} {(A' , B')} } (u , v) → record
5     { _{$}_ = λ {⟦ f , φ ⟧ → ⟦ v ∘ f ∘ u , (nHom u ∘r Cod) ∘v φ ⟧ }
6     ; cong = λ { ⟦ f , φ ⟧ } { ⟦ g , φ' ⟧ } (f≈g , φ≈φ') →
7       (∘-resp≈ Equiv.refl (∘-resp≈ f≈g Equiv.refl))
8       , (λ {x} → ∘-resp≈r (φ≈φ' {x}))
9     }
10 }
```

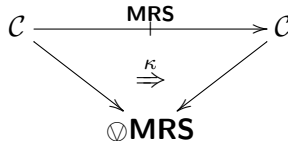
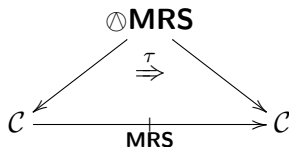
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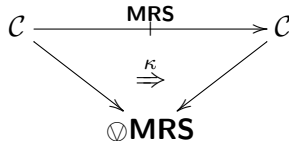
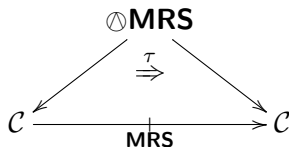
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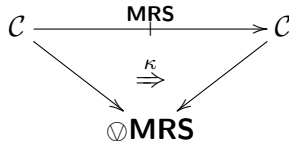
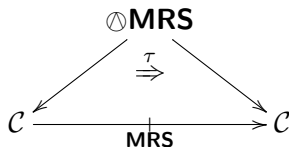


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- The **tabulator** and cotabulator: the universal categories with cells



- The category of **elements** $\mathcal{E}(\mathbf{MRS})$ via pullback;

The two-sided discrete fibration $\mathcal{C} \leftarrow \bigoplus \mathbf{MRS} \rightarrow \mathcal{C}$ is the Rosen fibration of the title.

Rosen fibration(s)

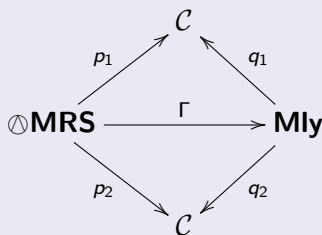
Arbib¹ showed each MR system determines a **Mealy automaton** (input A , output B , state $[A, B]$), functorially:

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Arbib¹ showed each MR system determines a **Mealy automaton** (input A , output B , state $[A, B]$), functorially:

There is a total category of Mealy automata and a fibred functor



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Rosen fibration(s)

Fibration of repair maps

The presheaf $Rep : \mathcal{C}^{op} \rightarrow \mathbf{Set}$, $A \mapsto \mathbf{Cat}(\mathcal{C}^{[0 \rightarrow 1]}, \mathcal{C})(cod, [A, cod])$ has the **fibration of repairs** $\mathcal{E}(Rep)$ as its category of elements.

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Objects: $(A, \Phi : cod \Rightarrow [A, cod])$; **morphisms** $u : (A, \Phi) \rightarrow (A', \Upsilon)$ making $\Phi = \Upsilon * [u, cod]$.

$$\begin{array}{ccc} & cod & \\ \Upsilon \swarrow & & \searrow \Phi \\ [A', cod] & \xrightarrow{[u, cod]} & [A, cod] \end{array}$$

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$$\begin{array}{ccc} & cod & \\ \Upsilon \swarrow & & \searrow \Phi \\ [A', cod] & \xrightarrow{[u, cod]} & [A, cod] \end{array}$$

The fibration of repairs is a **discrete, coreflective subfibration** of the tabulator:

$$\mathcal{E}(Rep) \begin{array}{c} \xleftarrow{k} \\ \xrightarrow{\mathbb{T}} \\ \mathbb{J} \end{array} \mathcal{T}(\mathbf{MRS}) \quad k(A, B; \xi @ (f, \Phi)) = (A, \Phi)$$

Rosen and others^{2 3 4}

²M. W. Warner, "Representations of (M,R) -systems by categories of automata". Bull. Math. Biology 44, 661–668 (1982).

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Rosen and others^{2 3 4} considered longer MR systems

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Rosen and others^{2 3 4} considered **longer** MR systems

$$A \longrightarrow B \longrightarrow [A, B] \longrightarrow [B, [A, B]] \longrightarrow \dots$$

defined by a **Fibonacci rule** $M_0 = A$,

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$$\begin{array}{ccccc} A & \longrightarrow & B & \xrightarrow{R(f,\Phi)} & [A, B] \\ & & \parallel & & \parallel \\ & & X & \xrightarrow{M(g,\Psi)} & Y \longrightarrow [X, Y] \end{array}$$

Longer MRs

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This 'vindicates Rosen': it makes *precise*, in *modern terms*, an idea he had decades ago.

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Thanks!

Speculations on the big picture

- Using structural mathematics to study “living” systems is an old idea;
- It remains unclear whether biology even admits a foundational treatment in terms of axioms and theorems (Schiaparelli, Freguglia. . .);
- “Living” systems interact with their environment through dual feedback loops, and exhibit self-repair, energy storage, resilience, and many other functions;
- *While category-theoretic frameworks exist for modeling feedback, resource theories, and other aspects individually, there is currently no framework capable of integrating all these dimensions into a single mathematical theory.*
- Any non-trivial contribution to this problem should be regarded as a valuable advance for **both mathematics and the life sciences**;
- To me, as a mathematician, one of Nature’s purposes is to inspire profound mathematical questions —a perspective that is often overlooked.